BUS13-4

Microeconomics

Choice under constraint, market equilibrium, and the limits of markets

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Prices coordinate the choices of strangers — until incentives, power, or information get in the way.

Albert School — 22 hours

Preface

This book is the entry point to economic reasoning at Albert. It assumes no prior economics and no mathematics beyond secondary-school algebra: you should be comfortable solving a linear equation, reading the slope of a straight line, and substituting numbers into a formula. Everything specific to economics — opportunity cost, marginal thinking, equilibrium, surplus, elasticity — is built from scratch.

Microeconomics studies **individual** decisions: a consumer with a fixed budget, a firm choosing how much to produce, two rivals setting prices. Its central claim is that these scattered choices are **coordinated** by prices, and that studying that coordination tells us when markets deliver good outcomes and when they fail. We will take rational choice seriously as a modelling device, and equally seriously the **behavioral** evidence that real people are boundedly rational, loss-averse, and present-biased — because knowing where the rational model breaks is part of using it well.

The book closes with a short bridge to national accounting (GDP). That is not macroeconomics; it is the **measurement literacy** the macro course (BUS23-4) will assume on day one. Read actively: every chapter opens with a concrete example before any general statement, flags the mistakes students reliably make, and ends with exercises. Do the exercises.

Contents

1 The economic way of thinking

1.1 Scarcity, choice, and opportunity cost

Start with a concrete afternoon. You have four free hours before a deadline. You can revise for an exam, take a paid shift worth €60, or see friends. You cannot do all three: time is **scarce**. Whatever you choose, the most valuable thing you gave up is the real cost of your choice.

Definition 1 (Opportunity cost) The **opportunity cost** of a decision is the value of the best alternative you forgo by making it. It is measured not in money spent but in **the next-best option lost**.

If you take the shift, the €60 is an explicit gain but the opportunity cost is the foregone revision (and the friends). Economics insists that the relevant cost of any action is what that action **displaces**, which is often not a cash outlay at all.

> **Worked example — The free ticket.** A friend gives you a free ticket to a concert tonight. The ticket has a market price of €50, but you would not have bought one. Tonight's alternative is a €0 evening at home that you value at €20 of enjoyment. The opportunity cost of attending is €20 — the enjoyment forgone — **not** the €50 face value, because you never would have paid it. "Free" things are rarely free; they cost whatever they displace.

Common pitfall Opportunity cost is **forward-looking**. Money or time already spent and unrecoverable is a **sunk cost** and is irrelevant to the choice in front of you (Section 1.4). The most common beginner error is to count cash outlays as costs while ignoring forgone alternatives that involve no cash.

1.2 Marginal reasoning

Economic decisions are rarely all-or-nothing. The right question is almost never "produce or not?" but "produce **one more** unit or not?" — a comparison at the **margin**.

Definition 2 (Marginal quantity) A **marginal** value is the change in a total caused by one more unit of activity. **Marginal benefit** (MB) is the extra benefit from one more unit; **marginal cost** (MC) is the extra cost. Keep doing the activity while $\text{MB} > \text{MC}$; stop where $\text{MB} = \text{MC}$.

> **Worked example — How many coffees to brew.** A café's marginal cost of one more espresso is €0.40 (beans, milk, energy). At the current price each sells for €0.90, so $\text{MB} = 0.90 > 0.40 = \text{MC}$: brew more. As the morning rush fades, the next customer would only pay €0.30; now $\text{MB} = 0.30 < 0.40$, so stop. The café maximizes its gain at the cup where $\text{MB} = \text{MC}$, regardless of how high its rent or how much it paid for the machine — those do not change with one more cup.

This "compare at the margin" logic recurs in every chapter: a consumer balances the marginal utility per euro across goods (Chapter 6), a firm sets $\text{MC} = \text{MR}$ (Chapter 8), a regulator sets a tax equal to marginal external damage (Chapter 10).

1.3 Incentives

Definition 3 (Incentive) An **incentive** is anything that changes the marginal benefit or marginal cost of an action, and so changes behaviour. Rational actors respond to incentives at the margin, often in ways the designer did not intend.

Worked example — The day-care fine. A day-care fines parents who collect children late. Lateness **rises**: the fine reprices a vague social obligation as a small, payable fee, lowering the moral cost of being late. The incentive worked — just not in the intended direction.

1.4 Positive vs normative, and sunk costs

Definition 4 (Positive vs normative claims) A **positive** claim describes what **is** and can in principle be tested ("a €1 tax raises the price by €0.60"). A **normative** claim asserts what **ought** to be and rests on values ("the tax is unfair"). Economics can settle positive disputes with evidence; normative disputes require a value judgement it cannot supply.

Definition 5 (Sunk cost) A **sunk cost** is an expenditure already incurred and unrecoverable. Because it is identical across every option still open, it is irrelevant to any forward-looking decision.

Worked example — Walking out of a bad film. You paid €12 for a cinema ticket; thirty minutes in, the film is dreadful. The €12 is gone whether you stay or leave — it is sunk. The only live comparison is the next 90 minutes of boredom versus 90 minutes doing something you enjoy. Staying "to get your money's worth" throws good time after sunk money.

Common pitfall The **sunk-cost fallacy** — continuing a project because of what has already been spent — is one of the most expensive mistakes in business. "We've invested €2M, we can't stop now" is exactly backwards: the €2M is gone; only future costs and benefits should decide whether to continue.

1.5 The production possibility frontier

The PPF makes scarcity, trade-offs, and opportunity cost visible. Consider an economy producing only two goods — say **bread** and **phones** — with fixed resources and technology.

Definition 6 (Production possibility frontier (PPF)) The **PPF** is the set of maximum output combinations of two goods attainable with given resources and technology. Points **on** the curve are efficient; points **inside** are attainable but wasteful; points **outside** are currently unattainable.

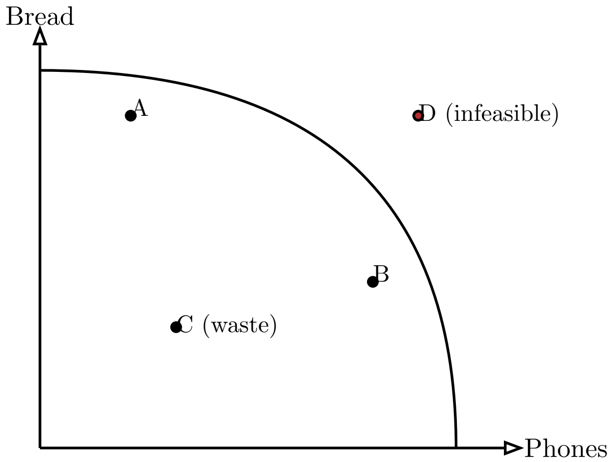

Figure 1: A production possibility frontier. Moving from A to B converts bread into phones; the bread given up is the opportunity cost of the extra phones. The outward bow reflects **increasing** opportunity cost.

The frontier bows outward because resources are **not** equally good at both tasks. The first phones are made by shifting the workers least suited to baking, costing little bread; squeezing out the last phones forces even master bakers onto the assembly line, costing a great deal of bread. Hence opportunity cost **rises** along the frontier — the slope steepens.

Chapter 1 in one line. Every choice has an opportunity cost (the best forgone alternative); decide at the margin (MB vs MC); ignore sunk costs; and keep positive ("what is") separate from normative ("what ought").

Exercises

1. You can spend Saturday on a €90 paid shift, on revision you value at €70, or at a festival you value at €120 (ticket already bought, non-refundable, €40). What is the opportunity cost of going to the festival? Should you go?

2. A firm spent €500{,}000 developing a product. Finishing it costs another €200{,}000 and would bring in €300{,}000 of revenue. Should the firm finish? Identify the sunk cost and the marginal comparison.

3. Classify each as positive or normative: (a) "Rent control reduces the supply of flats." (b) "Rent control is good for cities." (c) "Raising the carbon tax cuts emissions." (d) "The rich should pay more tax."

4. Draw a PPF for "study hours" vs "leisure hours" in a fixed day. Why is **this** frontier a straight line rather than bowed out?

2 Bounded rationality and behavioral realism

2.1 The rationality hypothesis and its limits

The models in the rest of this book lean on a working assumption.

> **Definition 7 (Rationality hypothesis)** A **rational** agent has stable, consistent preferences and chooses the available option that best satisfies them given their information and constraints. The assumption is a **modelling device**: it makes behaviour predictable, not a claim that people are flawless calculators.

The hypothesis earns its keep — supply and demand work remarkably well — but it fails in **systematic**, repeatable ways. Systematic failure matters because it can be predicted, designed around, and sometimes exploited. We review four findings.

2.2 Bounded rationality

> **Definition 8 (Bounded rationality)** Real agents have limited attention, memory, and computing power. Instead of optimizing, they **satisfice**: they use rules of thumb (heuristics) and stop at the first option that is good enough.

> **Worked example — Default options.** In countries where organ donation is opt-**out** (you are a donor unless you tick a box), consent rates exceed 90%; in otherwise similar opt-**in** countries they sit below 30%. A fully rational agent's deep preference should not depend on the default — but a boundedly rational one rarely overrides it. Defaults are therefore among the most powerful tools in policy and product design.

2.3 Framing and representativeness

> **Definition 9 (Framing effect)** Choices depend on how logically equivalent options are **described**. "90% survival" and "10% mortality" carry the same facts but pull decisions in opposite directions.

> **Definition 10 (Representativeness heuristic)** People judge probability by **similarity to a stereotype** rather than by base rates, producing predictable errors — e.g. assuming a quiet, tidy person is "more likely a librarian than a salesperson," ignoring that salespeople vastly outnumber librarians.

2.4 Loss aversion and present bias

> **Definition 11 (Loss aversion)** A loss looms larger than an equal gain: losing €100 hurts roughly twice as much as gaining €100 pleases. This makes people cling to what they own (the **endowment effect**) and avoid fair gambles.

> **Definition 12 (Present bias)** People over-weight immediate rewards relative to future ones, reversing their own plans as the moment arrives — preferring €100 today to €110 next

week, yet €110 in a year to €100 in 51 weeks. The hallmark is **preference reversal**: today's self and next-month's self disagree.

Worked example — Designing a savings plan. Present bias predicts people will **intend** to save and then not. "Save More Tomorrow" plans exploit this: employees commit **in advance** to route part of **future** raises into savings. Because the cost lands later, present bias no longer blocks it, and enrolment and saving rates jump.

Common pitfall Behavioral economics is about **incentive design** — defaults, commitment devices, framing of choices — and is distinct from the consumer-psychology framing used in marketing to drive sales. The mechanisms overlap (both use framing); the **purpose** and the welfare question ("is the chooser better off?") differ. Do not conflate "nudging people to save" with "nudging people to buy."

Chapter 2 in one line. The rational model is a useful default whose failures are **systematic**: bounded rationality (satisficing, defaults), framing and representativeness, loss aversion, and present bias — each a lever for designing incentives, not merely a curiosity.

Exercises

5. A gym sells a €600 annual membership and a €60 monthly pass. A member who visits 8 times a month would save with the annual plan but chooses monthly. Which behavioral finding best explains this, and how would you redesign the offer?

6. Reframe "this surgery has a 5% mortality rate" to make patients more willing to consent, changing no facts. Name the effect.

7. Give a business example where loss aversion makes a **free trial that auto-converts to paid** more profitable than a cheaper upfront price.

8. Explain why an opt-out pension scheme can raise participation more cheaply than a matching subsidy, using bounded rationality.

3 Supply, demand, and market equilibrium

3.1 Demand

Definition 13 (Demand curve) The **demand curve** gives the quantity buyers wish to purchase at each price, holding all else fixed. It slopes downward: a higher price reduces quantity demanded. We write it algebraically as $Q_D = a - bP$ with $a, b > 0$.

A **movement along** the curve is a change in quantity demanded caused by the good's own price. A **shift** of the whole curve is a change in demand caused by something else — income, the price of related goods, tastes, expectations, the number of buyers.

Common pitfall "Change in quantity demanded" (move along the curve, caused by **own price**) and "change in demand" (shift of the curve, caused by **anything else**) are not interchangeable. Conflating them is the single most common exam error in this chapter.

3.2 Supply

Definition 14 (Supply curve) The **supply curve** gives the quantity sellers wish to sell at each price, all else fixed. It slopes upward: a higher price draws out more output. Algebraically $Q_S = c + dP$ with $d > 0$.

3.3 Equilibrium

Definition 15 (Market equilibrium) **Equilibrium** is the price P^* at which $Q_D = Q_S$. At any higher price there is a **surplus** (sellers undercut each other, pushing price down); at any lower price a **shortage** (buyers bid the price up). Only at P^* is there no pressure to move.

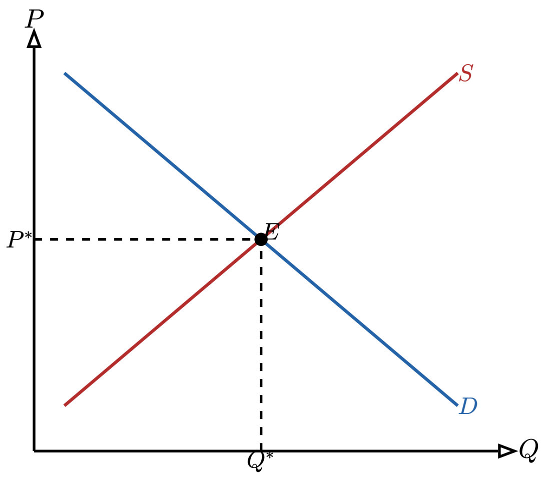

Figure 2: Market equilibrium E at the intersection of supply and demand. Above P^* supply exceeds demand (surplus); below it demand exceeds supply (shortage).

Worked example — Solving for equilibrium. Let $Q_D = 120 - 2P$ and $Q_S = -30 + 3P$ (quantities in thousands, price in €). Set $Q_D = Q_S$:

$$120 - 2P = -30 + 3P \quad \rightarrow \quad 150 = 5P \quad \rightarrow \quad P^* = 30.$$

Substitute back: $Q^* = 120 - 2(30) = 60$ thousand units. Always check with the other equation: $-30 + 3(30) = 60$. ✓

3.4 Comparative statics: shocks

Definition 16 (Demand / supply shock) A **shock** shifts a curve. A **demand** increase (rightward shift) raises both P^* and Q^*. A **supply** increase (rightward shift) lowers P^* but raises Q^*. Read off the new equilibrium by re-solving with the shifted equation.

Worked example — A supply shock. A frost destroys part of the coffee crop, shifting supply to $Q_S = -60 + 3P$ (30 fewer thousand at every price). With demand unchanged:

$$120 - 2P = -60 + 3P \quad \to \quad 180 = 5P \quad \to \quad P^* = 36,$$

and $Q^* = 120 - 2(36) = 48$. The frost raised the price from €30 to €36 and cut quantity from 60 to 48 thousand — exactly the upward-price, lower-quantity pattern a leftward supply shift predicts.

Common pitfall When **both** curves shift, only one of price or quantity has a determinate direction; the other depends on the relative sizes of the shifts. Draw it — do not guess.

Chapter 3 in one line. Demand slopes down, supply slopes up, equilibrium sets $Q_D = Q_S$; solve it by equating the two lines, and trace any shock by re-solving with the shifted curve.

Exercises

9. With $Q_D = 200 - 4P$ and $Q_S = 20 + 2P$, find P^* and Q^*.
10. Income rises and demand becomes $Q_D = 260 - 4P$ (supply unchanged). Find the new equilibrium and describe the direction of change.
11. A new technology shifts supply to $Q_S = 80 + 2P$ (original demand). Find the new equilibrium. Did price and quantity move as theory predicts?
12. Explain, with a diagram, why a price **ceiling** set below P^* creates a persistent shortage while a ceiling above P^* does nothing.

4 Elasticity

4.1 Price elasticity of demand

How **responsive** is quantity to price? "A lot" or "a little" is not enough; we need a unit-free number so we can compare petrol with restaurant meals.

Definition 17 (Price elasticity of demand)

$$\varepsilon_d = \frac{\%\text{ change in } Q_D}{\%\text{ change in } P} = \frac{\Delta Q/Q}{\Delta P/P}.$$

Because demand slopes down, ε_d is negative; we usually quote its absolute value. Demand is **elastic** if $|\varepsilon_d| > 1$, **inelastic** if $|\varepsilon_d| < 1$, **unit-elastic** if $|\varepsilon_d| = 1$.

Worked example — Computing an elasticity. Price rises from €20 to €22 (a +10% change) and quantity falls from 100 to 85 (a -15% change). Then $\varepsilon_d = -15\%/+10\% = -1.5$. In absolute value $1.5 > 1$: demand is elastic, so buyers are quite responsive here.

Common pitfall Elasticity is **not** the slope. The slope $\Delta Q/\Delta P$ has units and is constant along a straight-line demand; elasticity is unit-free and **varies** along that same line — elastic near the top, inelastic near the bottom. Two different ideas.

4.2 Elasticity and total revenue

This is the result managers care about.

Proposition 18 (Elasticity–revenue rule) Total revenue is $\text{TR} = P \times Q$. Raising price:

- **raises** TR when demand is inelastic ($|\varepsilon_d| < 1$): the price gain beats the quantity loss;
- **lowers** TR when demand is elastic ($|\varepsilon_d| > 1$): the quantity loss dominates;
- leaves TR unchanged at unit elasticity.

Worked example — Should the museum raise admission?. A museum estimates $|\varepsilon_d| = 0.4$ for tickets (inelastic — few substitutes for a special exhibit). Raising the price 10% cuts attendance only about 4%, so TR rises roughly $10\% - 4\% = 6\%$. Raising price is the revenue-maximizing move **here**. For a downtown café with $|\varepsilon_d| = 2.5$, the same 10% rise would cut sales 25% and shrink revenue — a cut might serve it better.

4.3 Income and cross-price elasticities

Definition 19 (Income elasticity)

$$\varepsilon_I = \frac{\%\text{ change in } Q}{\%\text{ change in income}}.$$

$\varepsilon_I > 0$ marks a **normal** good; $\varepsilon_I < 0$ an **inferior** good (bought less as income rises, e.g. bus travel); $\varepsilon_I > 1$ a **luxury**.

Definition 20 (Cross-price elasticity)

$$\varepsilon_{XY} = \frac{\%\text{ change in } Q_X}{\%\text{ change in } P_Y}.$$

$\varepsilon_{XY} > 0$ means X and Y are **substitutes** (butter and margarine); $\varepsilon_{XY} < 0$ means **complements** (printers and ink).

Worked example — Reading the sign. When the price of train tickets rises 10%, airline bookings on the same route rise 6%: $\varepsilon_{XY} = +6\%/+10\% = +0.6 > 0$, so trains and flights are substitutes. When fuel prices rise 10% and SUV sales fall 8%, $\varepsilon_{XY} = -8\%/+10\% = -0.8 < 0$: fuel and SUVs are complements.

Chapter 4 in one line. Elasticity is the unit-free responsiveness of quantity; its **sign** classifies goods (normal/inferior, substitute/complement) and its **size** tells you whether a price rise grows or shrinks revenue.

Exercises

13. Price falls from €50 to €45 and quantity rises from 200 to 260. Compute ε_d and classify demand.

14. A streaming service with $|\varepsilon_d| = 0.7$ wants more revenue. Raise or cut the price? By the elasticity rule, justify in one sentence.

15. A 5% income rise raises restaurant spending 9% and cuts instant-noodle sales 4%. Compute both income elasticities and classify each good.

16. Explain why the same straight-line demand curve is elastic at high prices and inelastic at low prices, referring to the $\Delta Q/Q$ and $\Delta P/P$ terms.

5 Welfare: surplus, taxes, and deadweight loss

5.1 Consumer and producer surplus

Equilibrium is not just a price — it creates **value**. Surplus measures it.

Definition 21 (Consumer surplus (CS)) **Consumer surplus** is the gap between what buyers were willing to pay (the height of the demand curve) and what they actually paid, summed over all units. On a diagram it is the triangle **below demand and above price**.

Definition 22 (Producer surplus (PS)) **Producer surplus** is the gap between the price sellers received and the minimum they would have accepted (the height of the supply curve), summed over all units — the triangle **above supply and below price**. Total surplus TS = CS + PS measures the market's total gains from trade.

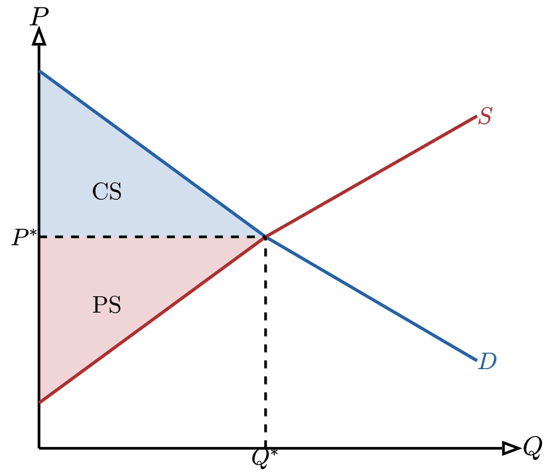

Figure 3: Consumer surplus (blue) sits below demand and above P^*; producer surplus (red) sits above supply and below P^*. Their sum is the gains from trade at the competitive equilibrium.

Worked example — Surplus as a triangle area. Suppose at equilibrium $P^* = 30$, $Q^* = 60$, the demand intercept (highest willingness to pay) is €60, and the supply intercept (lowest acceptable price) is €0. Then

$$\mathrm{CS} = \frac{1}{2} \times 60 \times (60 - 30) = 900, \quad \mathrm{PS} = \frac{1}{2} \times 60 \times (30 - 0) = 900,$$

so total surplus is €1{,}800 (with Q in the same units). Each triangle is $\frac{1}{2} \times \text{base} \times \text{height}$ — base is the equilibrium quantity, height is the price gap at $Q = 0$.

5.2 Tax incidence and deadweight loss

Now put a per-unit tax t on the good. The price buyers pay and the price sellers keep split apart by exactly t.

Definition 23 (Tax incidence) **Incidence** is who actually bears a tax, regardless of who legally pays it. A tax t drives a wedge between the buyer's price P_b and the seller's price P_s with $P_b - P_s = t$. Trade falls from Q^* to Q_t. The side that is **less elastic** bears the larger share — it has fewer alternatives, so it cannot dodge the tax.

Definition 24 (Deadweight loss (DWL)) Some mutually beneficial trades (those between Q_t and Q^*) no longer happen. The surplus on these lost trades goes to **no one** — not to consumers, producers, or the government. This pure waste is the **deadweight loss**, the triangle between supply and demand from Q_t to Q^*.

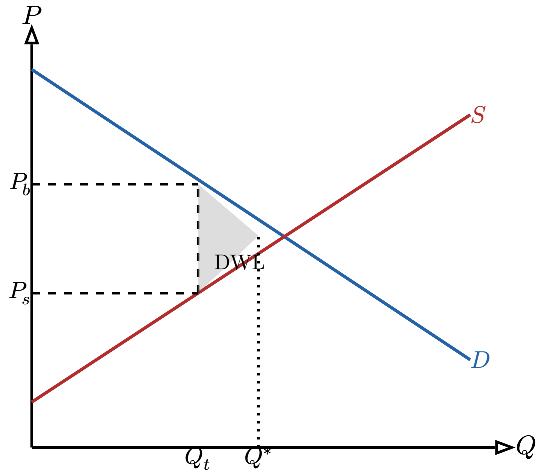

Figure 4: A per-unit tax opens a wedge $P_b - P_s = t$, cutting traded quantity to Q_t. Government revenue is the rectangle $t \times Q_t$; the grey triangle between Q_t and Q^* is the deadweight loss — surplus destroyed, collected by no one.

Worked example — Computing incidence and DWL. Demand $Q_D = 120 - 2P$, supply $Q_S = -30 + 3P$; pre-tax $P^* = 30$, $Q^* = 60$ (Chapter 3). Impose $t = 5$ per unit, paid by sellers, so sellers receive $P - 5$. Set $Q_D = Q_S$ at the buyer price P_b with $P_s = P_b - 5$:

$$120 - 2P_b = -30 + 3(P_b - 5) \quad \to \quad 120 - 2P_b = -45 + 3P_b \quad \to \quad 165 = 5P_b \quad \to \quad P_b = 33.$$

So $P_s = 28$ and $Q_t = 120 - 2(33) = 54$. Buyers pay €3 more (33 vs 30), sellers keep €2 less (28 vs 30): the €5 tax splits 3:2. Sellers are more responsive here ($d = 3 > b = 2$), so buyers — the less elastic side — bear the larger share. Revenue $= t \times Q_t = 5 \times 54 = 270$. Deadweight loss $= \frac{1}{2} \times t \times (Q^* - Q_t) = \frac{1}{2} \times 5 \times (60 - 54) = 15$.

Common pitfall Who **writes the cheque** to the tax office does not determine who **bears** the tax. In the example the legal burden was on sellers, yet buyers bore €3 of the €5. Incidence is set by **elasticities**, not by the statute.

Chapter 5 in one line. Markets generate surplus (CS + PS = the gains from trade); a tax transfers some surplus to the government but destroys a deadweight triangle, and its real burden falls on the **less elastic** side.

Exercises

17. At equilibrium $P^* = 25$, $Q^* = 100$, demand intercept €75, supply intercept €0. Compute CS, PS, and total surplus.

18. With $Q_D = 200 - 4P$ and $Q_S = 20 + 2P$, impose a €6 per-unit tax on buyers. Find P_b, P_s, Q_t, government revenue, and the deadweight loss.

19. In the previous exercise, what fraction of the tax do buyers bear? Relate your answer to the relative slopes of supply and demand.
20. Explain why a tax on a good with **perfectly inelastic** demand raises revenue with **zero** deadweight loss. Draw it.

6 Consumer theory

6.1 Utility and indifference curves

We now open the black box behind the demand curve: how a consumer with a fixed budget chooses.

Definition 25 (Utility and indifference curves) **Utility** ranks bundles of goods by preference. An **indifference curve** connects all bundles giving the **same** utility — the consumer is indifferent among them. Curves farther from the origin are preferred (more is better); they never cross; and they are **convex** to the origin, reflecting a taste for balanced bundles.

Definition 26 (Marginal rate of substitution (MRS)) The **MRS** is the slope of an indifference curve: the amount of good Y the consumer will give up for one more unit of good X while staying equally happy. Convexity means the MRS **diminishes** — the more X you already have, the less Y you will sacrifice for further X.

6.2 The budget constraint and optimum

Definition 27 (Budget constraint) With income I and prices P_X, P_Y, affordable bundles satisfy $P_X X + P_Y Y \leq I$. The boundary is the **budget line**, with slope $-P_X/P_Y$ — the rate at which the **market** lets you trade Y for X.

The consumer's best affordable bundle is where an indifference curve just **touches** the budget line.

Proposition 28 (Consumer optimum) At the optimal bundle the indifference curve is tangent to the budget line:

$$\mathrm{MRS} = \frac{P_X}{P_Y}, \quad \text{equivalently} \quad \frac{(\mathrm{MU})_X}{P_X} = \frac{(\mathrm{MU})_Y}{P_Y}.$$

The personal trade-off rate equals the market trade-off rate; equivalently, the last euro spent on each good buys the same marginal utility. If they differ, shifting spending toward the higher "bang per buck" good raises utility.

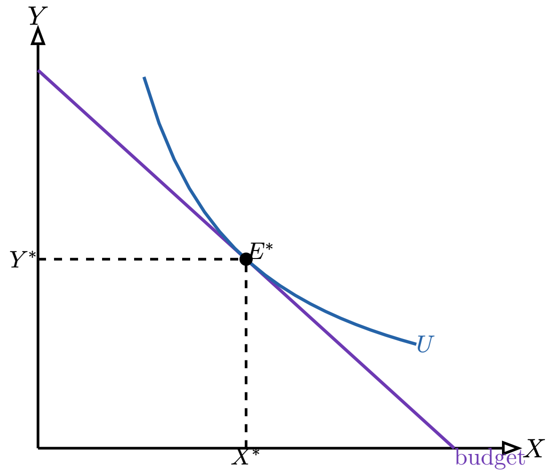

Figure 5: The optimum E^* is where the highest reachable indifference curve is tangent to the budget line: $\text{MRS} = P_X/P_Y$.

6.3 Income and substitution effects

When a price changes, the response splits into two conceptually distinct pieces — a decomposition that matters for pricing and policy.

Definition 29 (Substitution and income effects) When P_X falls, the consumer buys more X for two reasons:

- the **substitution effect** — X is now cheaper **relative to** Y, so they swap toward X even at unchanged real income (always **increases** Q_X);
- the **income effect** — the price cut makes them effectively richer, raising purchasing power; this raises Q_X for a normal good but **lowers** it for an inferior good.

Total effect = substitution + income.

Worked example — A price cut on a normal good. A subscription service halves its price. The **substitution** effect: it is now cheaper than rival entertainment, so subscribers switch toward it. The **income** effect: the saving leaves households with spare cash, some of which buys yet more of this (normal) service. Both push the same way, so quantity rises strongly — useful to know when forecasting how a price cut will lift volume.

Common pitfall For a **Giffen** good (a strongly inferior staple), the income effect can be large enough to **reverse** the substitution effect, so demand slopes **upward** over a range. Giffen goods are a rare curiosity — do not assume them — but they show why the decomposition matters: "price down $\Rightarrow$ quantity up" is the **normal** case, not a law.

Chapter 6 in one line. A consumer maximizes utility where an indifference curve is tangent to the budget line ($\text{MRS} = P_X/P_Y$); a price change moves them via a substitution effect (always toward the cheaper good) plus an income effect (sign depends on normal vs inferior).

Exercises

21. Income €120, $P_X = 12$, $P_Y = 6$. Write the budget line and its slope. How many Y if all income goes to Y?
22. At a bundle, $(\mathrm{MU})_X = 20$, $(\mathrm{MU})_Y = 5$, $P_X = 4$, $P_Y = 2$. Is the consumer optimizing? If not, which good should they buy more of?
23. The price of bus tickets (an inferior good for a commuter) falls. Decompose the effect into substitution and income parts; can total bus rides fall?
24. Draw two indifference curves and explain geometrically why they cannot cross, using transitivity of preferences.

7 Firm theory: production and costs

7.1 Production and diminishing returns

Definition 30 (Production function) A **production function** $Q = f(L)$ gives output from inputs (here labour L, capital fixed in the short run). The **marginal product** of labour, MP_L, is the extra output from one more worker.

Definition 31 (Diminishing marginal returns) As more of a variable input is added to fixed inputs, the marginal product eventually **falls**: the tenth cook in one kitchen adds less than the third. This is the engine behind rising marginal cost.

7.2 The cost family

Definition 32 (Fixed, variable, total, average, marginal cost) **Fixed cost** (FC) does not vary with output (rent); **variable cost** (VC) does (materials, wages). Then

$$\mathrm{TC} = \mathrm{FC} + \mathrm{VC}, \quad \mathrm{ATC} = \frac{\mathrm{TC}}{Q}, \quad \mathrm{AVC} = \frac{\mathrm{VC}}{Q}, \quad \mathrm{MC} = \frac{\Delta\,\mathrm{TC}}{\Delta Q}.$$

MC is the cost of one more unit; ATC is cost **per** unit.

Worked example — Building the cost table. Fixed cost €100. Output and variable cost:

Q	VC	TC	MC	AVC	ATC
0	0	100	—	—	—
1	50	150	50	50.0	150.0
2	90	190	40	45.0	95.0
3	120	220	30	40.0	73.3
4	160	260	40	40.0	65.0
5	220	320	60	44.0	64.0
6	300	400	80	50.0	66.7

MC falls then rises (diminishing returns set in after $Q = 3$); ATC is U-shaped, bottoming near $Q = 5$. Notice MC cuts AVC and ATC at **their minima** — a fact we use next chapter.

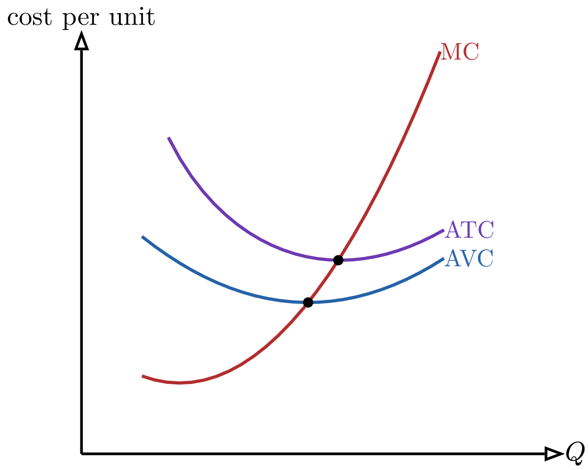

Figure 6: The short-run cost curves. MC is U-shaped and passes through the **minimum** of both AVC and ATC. The ATC–AVC gap is average fixed cost, shrinking as fixed cost spreads over more units.

7.3 Break-even and shutdown

Definition 33 (Break-even and shutdown points) The **break-even price** is the minimum of ATC: below it the firm makes a loss. The **shutdown price** is the minimum of AVC: below it the firm cannot even cover variable cost, so it loses **less** by producing nothing. Between the two, the firm keeps operating in the short run to defray fixed costs it must pay regardless.

Common pitfall Shutdown compares price with **average variable** cost, not **average total** cost. A firm losing money can still be right to keep producing, as long as price covers AVC — every euro above variable cost chips away at the fixed cost it owes either way. The fixed cost is, in the short run, sunk (Chapter 1).

Chapter 7 in one line. Diminishing returns make MC rise; from TC we get the U-shaped ATC/AVC, MC cuts both at their minima, and a firm produces while price $\geq$ min AVC (shutdown) even if price $<$ min ATC (loss).

Exercises

25. With FC $=$ €80 and VC at $Q = 1, 2, 3, 4$ of €40, €70, €110, €170, build the TC, MC, AVC, and ATC columns.

26. From your table, identify the output that minimizes ATC and the one that minimizes AVC.

27. If the market price is €45, should this firm produce? At €25? Use the shutdown rule.

28. Explain why MC must intersect ATC at ATC's minimum (hint: when the marginal is below the average, the average falls).

8 Competition: profit maximization and the zero-profit theorem

8.1 The profit-maximizing rule

Definition 34 (Marginal revenue (MR)) **Marginal revenue** is the extra revenue from selling one more unit. For a **price-taking** firm in perfect competition, each unit sells at the market price, so $\mathrm{MR} = P$.

Proposition 35 (Profit maximization: MC = MR) A firm maximizes profit at the output where $\mathrm{MC} = \mathrm{MR}$ (and MC is rising). If $\mathrm{MR} > \mathrm{MC}$, one more unit adds more to revenue than to cost — expand. If $\mathrm{MR} < \mathrm{MC}$, the last unit lost money — contract. Only $\mathrm{MC} = \mathrm{MR}$ leaves no improving move. For a competitive firm this is $\mathrm{MC} = P$.

Worked example — Choosing output as a price-taker. Take the cost table from Chapter 7 (FC €100). The market price is €60, so $\mathrm{MR} = 60$. MC reaches 60 at $Q = 5$ (MC: 50, 40, 30, 40, 60, …). Produce 5 units. Profit $= \mathrm{TR} - \mathrm{TC} = 60 \times 5 - 320 = 300 - 320 = -20$. The firm loses €20 — but price €60 exceeds min AVC (≈€40), so it produces anyway rather than lose the full €100 fixed cost.

8.2 Perfect competition in the long run

Definition 36 (Perfect competition) Many small firms sell an identical product; buyers and sellers are price-takers; and there is **free entry and exit**. Each firm faces a horizontal demand line at the market price.

Proposition 37 (Zero-profit theorem) Under perfect competition, long-run economic profit is driven to **zero**. Positive profit attracts entrants, raising supply and pushing price down; losses drive firms out, cutting supply and pushing price up. Entry and exit stop only when price equals minimum ATC, so $P = \mathrm{MC} = \min \mathrm{ATC}$ and economic profit is zero.

Common pitfall "Zero economic profit" is not "zero accounting profit." Economic profit already subtracts the **opportunity cost** of the owner's capital and labour (Chapter 1). Zero economic profit means the firm earns exactly its next-best return — a perfectly healthy business, not a failing one.

8.3 Monopolistic competition

Definition 38 (Monopolistic competition) Many firms sell **differentiated** products (cafés, hairdressers, brands of cereal). Each has slight pricing power — a downward-sloping demand — but free entry still drives long-run profit to zero. The outcome: $P > \mathrm{MC}$ (a markup) yet zero profit, with firms producing below minimum-ATC scale (some excess capacity) — the price of variety.

Chapter 8 in one line. Every firm maximizes profit at $\text{MC} = \text{MR}$; for a competitive price-taker that is $\text{MC} = P$, and free entry/exit then grinds long-run economic profit to zero at $P = \min \text{ATC}$.

Exercises

29. Using the Chapter 7 table, find the profit-maximizing output and profit if $P = 80$.

30. At $P = 64$ ($\approx$ min ATC for that table), what is the firm's profit? Relate to the zero-profit theorem.

31. Explain, step by step via entry, why a competitive industry earning positive profit must see price fall over time.

32. Why does a monopolistically competitive firm produce where $P > \text{MC}$ yet still make zero long-run profit? What does society get in exchange for the markup?

9 Market power: monopoly, price discrimination, and oligopoly

9.1 Monopoly

Definition 39 (Monopoly) A **monopolist** is the sole seller and faces the whole downward-sloping market demand. To sell more it must **lower the price on every unit**, so marginal revenue lies **below** the demand curve: MR $< P$. It still maximizes at MC $=$ MR, but then charges the higher price the demand curve allows at that quantity.

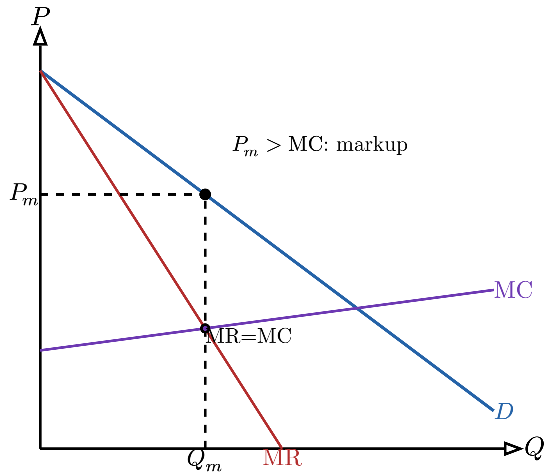

Figure 7: Monopoly. Output is set where MR $=$ MC (Q_m), but price is read **up** on the demand curve (P_m). Because $P_m >$ MC, the monopolist restricts output below the competitive level, creating a deadweight loss.

Worked example — Monopoly output, price, and DWL. Demand $P = 100 - Q$, so MR $= 100 - 2Q$ (same intercept, twice the slope). Constant MC $= 20$. Set MR $=$ MC: $100 - 2Q = 20 \to Q_m = 40$, and $P_m = 100 - 40 = 60$. A competitive market would produce where $P =$ MC: $100 - Q = 20 \to Q_c = 80$ at $P = 20$. The monopolist sells **half** as much at **triple** the price. Deadweight loss is the triangle between demand and MC from Q_m to Q_c: $\frac{1}{2} \times (60 - 20) \times (80 - 40) = \frac{1}{2} \times 40 \times 40 = 800$.

9.2 Price discrimination

Definition 40 (Price discrimination) Charging different prices for the same good to capture more surplus.

- **First-degree** (perfect): each buyer pays their exact willingness to pay (bespoke negotiation, personalized pricing). Captures **all** surplus.
- **Second-degree**: price varies with **quantity or version** chosen by the buyer (bulk discounts, good/better/best tiers); buyers self-select.
- **Third-degree**: different **identifiable groups** pay different prices (student, senior, regional pricing), with the lower price going to the more price-elastic group.

Worked example — Why students pay less. Software priced at €200 for firms and €40 for students is third-degree discrimination: students have elastic demand (many free alternatives, tight budgets) and firms inelastic demand. Charging each group near its own revenue-maximizing price beats a single price — and, by serving students who would not pay €200, can even **raise** total output.

9.3 Oligopoly and strategic interaction

When a few firms dominate, each one's best move depends on its rivals'. This is the realm of **game theory**.

Definition 41 (Nash equilibrium) A **Nash equilibrium** is a combination of strategies in which no player can do better by **unilaterally** changing their own choice, given what the others do. To find one, check each cell: is either player tempted to deviate? If neither is, it is a Nash equilibrium.

Worked example — The prisoner's dilemma (a price war). Two airlines each choose **High** or **Low** fares. Payoffs are annual profits (€m), written (Row, Column):

	B: High	**B: Low**
A: High	(10, 10)	(2, 14)
A: Low	(14, 2)	(5, 5)

Whatever B does, A earns more by choosing **Low** ($14 > 10$ if B High; $5 > 2$ if B Low), so **Low** is a dominant strategy — and symmetrically for B. The unique Nash equilibrium is (**Low, Low**) with profits (5, 5), even though both would prefer (**High, High**) = (10, 10). Individually rational price-cutting makes both worse off: the logic behind real price wars, and why cartels are unstable.

Common pitfall The Nash outcome need not be the **jointly best** outcome. (Low, Low) is Nash; (High, High) gives both more but is not self-enforcing because each firm is tempted to defect. "Stable" $\neq$ "efficient."

Chapter 9 in one line. Market power means $P > \mathrm{MC}$: a monopolist restricts output where $\mathrm{MR} = \mathrm{MC}$ (deadweight loss), price discrimination captures extra surplus, and oligopolists land in a Nash equilibrium that can trap them below the jointly best outcome.

Exercises

33. Demand $P = 120 - 2Q$, constant $\mathrm{MC} = 20$. Find the monopoly Q_m, P_m, and the deadweight loss versus the competitive outcome.
34. Classify each: (a) a cinema's matinee discount; (b) a quantity discount on printer paper; (c) an auctioneer extracting each bidder's top bid. Which degree?
35. Two firms choose **Advertise** or **Not**. Payoffs (Row, Col): both Not (8,8); A advertises only (12,4); B only (4,12); both (6,6). Find the Nash equilibrium and say whether it is jointly best.
36. Explain why a monopolist never produces on the **inelastic** part of its demand curve (hint: there, cutting output raises revenue **and** lowers cost).

10 Market failures: externalities and asymmetric information

Competitive markets are efficient only under conditions that often fail. This chapter covers the two classic failures: costs that spill onto bystanders, and trades where one side knows more than the other.

10.1 Externalities and Pigouvian taxes

Definition 42 (Externality) An **externality** is a cost or benefit imposed on a third party not reflected in the price. A **negative** externality (pollution, congestion) means the **social** cost exceeds the **private** cost, so the free market **over**-produces. A **positive** externality (vaccination, education) means social benefit exceeds private benefit, so the market **under**-produces.

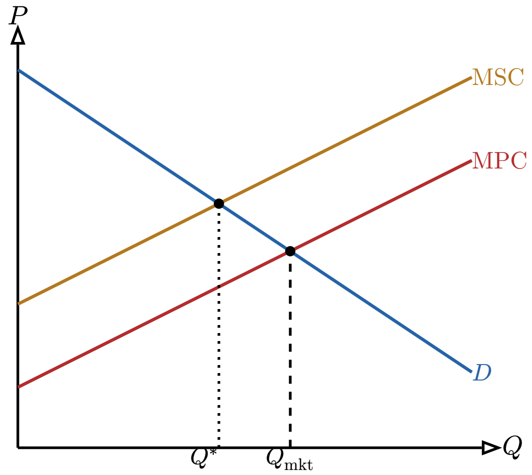

Figure 8: A negative externality. The market trades at Q_{mkt} where private cost (MPC) meets demand, but the social optimum Q^* — where **social** cost (MSC) meets demand — is lower. The gap is overproduction; the wedge between MSC and MPC is the external cost per unit.

Definition 43 (Pigouvian tax) A **Pigouvian tax** equal to the marginal external cost makes the producer face the full social cost: it shifts MPC up onto MSC, moving the market to the social optimum Q^*. The tax does not "distort" — it **corrects** a pre-existing distortion.

Worked example — Sizing a carbon tax. A plant emits 1 tonne of CO_2 per unit of output, and each tonne does €30 of external damage. Setting a Pigouvian tax of €30 per unit forces the firm to internalize that damage: its private marginal cost rises by exactly €30, so it cuts output to the level where buyers' willingness to pay covers the **full** social cost. The externality is internalized at the efficient quantity.

10.2 Asymmetric information

Definition 44 (Adverse selection (Akerlof's market for lemons)) When **sellers** know product quality but **buyers** do not, buyers will only pay the price of **average** quality. That price is too low for good ("peach") sellers, who exit, lowering average quality — which lowers the price further, and so on. Good products are driven out by bad ("lemons"); the market can unravel entirely. This is **adverse selection**: a hidden-quality problem present **before** the deal.

Worked example — Used cars. Half of used cars are peaches (worth €10{,}000) and half lemons (worth €4{,}000), but only sellers know which. A buyer offers the average, €7{,}000. At €7{,}000 no peach owner sells — their car is worth more — so only lemons remain, and a rational buyer revises down to €4{,}000. The good cars never trade. Warranties, certification, and brand reputation exist to **signal** quality and reopen the market.

Definition 45 (Moral hazard) **Moral hazard** is a **hidden-action** problem arising **after** a deal: once insured or paid up front, a party takes more risk or less care because they no longer bear the full consequences. An insured driver locks the car less; a salaried worker may slacken. Remedies align incentives: deductibles, co-payments, performance pay, monitoring.

Common pitfall Adverse selection and moral hazard both stem from asymmetric information but differ in **timing**: adverse selection is **hidden information before** the contract (you cannot tell a lemon from a peach); moral hazard is **hidden action after** it (you cannot watch the insured driver). Pick the remedy to match — screening and signalling for the former, incentive contracts and monitoring for the latter.

Chapter 10 in one line. Markets fail when costs spill onto third parties (externalities — fix with a Pigouvian tax equal to the marginal damage) or when one side knows more (adverse selection **before** the deal, moral hazard **after**), each with its own remedy.

Exercises

37. A factory's private MC is $20 + Q$ and each unit does €10 of external damage; demand is $P = 80 - Q$. Find the market quantity, the social optimum, and the Pigouvian tax that closes the gap.

38. Is each a positive or negative externality, and does the market over- or under-produce? (a) flu vaccination; (b) a noisy late-night bar; (c) a beekeeper next to an orchard.

39. Health insurance attracts disproportionately sick applicants and makes the insured less careful about their health. Identify which part is adverse selection and which is moral hazard, and propose one remedy for each.

40. Explain how a multi-year manufacturer warranty addresses adverse selection in the new-car market.

11 Bridge to macro: GDP and national accounts

This final chapter is not macroeconomics — it is the **measurement literacy** BUS23-4 will assume. To talk about growth, inflation, or central banking, you first need to know what "the size of the economy" means and how it is counted.

11.1 What GDP is

Definition 46 (Gross domestic product (GDP)) **GDP** is the market value of all **final** goods and services produced within a country in a period. Three words carry the weight: **final** (intermediate inputs are excluded to avoid double-counting), **produced** (not merely traded — second-hand sales and financial transfers do not count), and **within a country** (domestic, not by-nationality).

11.2 Three ways to measure the same total

Proposition 47 (The three approaches) Because every euro of output is also a euro of spending and a euro of income, GDP can be measured three equivalent ways:

- **Production**: sum the **value added** (output minus intermediate inputs) of every firm.
- **Expenditure**: $\text{GDP} = C + I + G + (X - M)$ — consumption, investment, government spending, plus net exports.
- **Income**: sum all wages, profits, rents, and interest earned in production.

In principle the three give the same number.

Worked example — Value added avoids double-counting. A farmer sells wheat to a miller for €1.00; the miller sells flour to a baker for €1.80; the baker sells bread for €3.00. Adding €1.00 + €1.80 + €3.00 = €5.80 would triple-count the wheat. The correct contribution to GDP is the **value added**: $1.00 + (1.80 - 1.00) + (3.00 - 1.80) = 1.00 + 0.80 + 1.20 = 3.00$ — exactly the price of the **final** good, the bread.

Common pitfall Many large flows are **not** in GDP: buying an existing house or a share of stock (no new production — only the estate-agent's or broker's **fee**, a service produced now, counts), and government **transfer** payments like pensions or unemployment benefits (money moved, nothing produced). The test is always: **was something new produced this period?**

11.3 What GDP misses

Definition 48 (Limits of GDP) GDP is a measure of **marketed production**, not of welfare. It omits, among others:

- **Non-market and household production** — unpaid childcare, cooking, volunteering — so the same activity counts only when paid for.
- **The distribution of income** — GDP can rise while most people are no better off.
- **Environmental depletion and externalities** — pollution that lowers wellbeing can **raise** GDP (clean-up is counted as production).
- **Leisure, health, and quality** — a longer working year raises GDP; more free time does not.

Worked example — When GDP and welfare part ways. A storm wrecks a coastline. Rebuilding raises GDP — construction, materials, wages all counted — yet the country is plainly worse off than before the storm. Likewise, if you marry your cook, measured GDP **falls** (the same meals are now unpaid) though nothing real changed. These are not flaws to be "fixed" so much as reminders that GDP answers one specific question: how much marketed output was produced?

Chapter 11 in one line. GDP is the market value of **final** output produced in a country, countable three equivalent ways ($C + I + G + (X - M)$, value added, or income); it is indispensable but blind to non-market work, distribution, the environment, and quality — the caveats BUS23-4 will carry forward.

Exercises

41. A logger sells timber to a furniture maker for €400; the maker sells tables for €1{,}000. What is the contribution to GDP, by the value-added method?

42. For each, in GDP or not, and why: (a) a new car bought by a household; (b) a second-hand car bought from a neighbour; (c) the dealer's commission on (b); (d) a state pension payment; (e) a haircut.

43. Given $C = 700$, $I = 200$, $G = 300$, exports $X = 150$, imports $M = 180$, compute GDP by the expenditure method.

44. Give two distinct reasons a country with higher GDP per capita than its neighbour might nonetheless have lower wellbeing.